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Sovereign AI and European Strategic Infrastructure

Compute, Energy, and the Political Economy of Europe's AI Continent Plan

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Prepared by SYSTEM INC AI Labs

Publication date

15 July 2026

Version

1.0 — Initial publication

Classification

Public — for distribution to institutional and government stakeholders

Suggested citation

BLACK& Research. 2026. "Sovereign AI and European Strategic Infrastructure." BLACK& Research Series, v1.0. Botoșani: BLACK HOLDINGS SRL.

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Executive Summary

Europe controls less than 5% of global AI compute capacity, while a handful of US hyperscalers account for roughly 70% of the region's cloud market. The European Union's response, the AI Continent Action Plan, rests on a network of 19 EuroHPC AI Factories already operating or selected, and a planned tier of up to five AI Gigafactories, each designed to house more than 100,000 advanced processors. The programme is backed by the InvestAI Facility, which aims to mobilise up to €200 billion in combined public and private capital, of which €20 billion is earmarked as public co-investment.

This paper examines the architecture, financing, and political economy of that build-out, and uses Romania's Black Sea AI Gigafactory proposal — a €4-5 billion, 1,500 MW facility spanning the Cernavodă nuclear site and the Doicești SMR project — as a detailed case study of how a mid-sized EU member state is positioning itself within the programme.

Three findings stand out. First, the earliest gigafactory-scale facilities on European soil are being built by commercial actors — Microsoft/Nscale in Portugal, Deutsche Telekom/NVIDIA in Germany, Mistral in France — ahead of the EU's own formal call for proposals, which has slipped from an original 2025 target into Q2 2026. Second, energy, not chip access, is emerging as the binding constraint on European AI infrastructure: European industrial electricity prices run roughly double US levels, and grid capacity — not GPU allocation — increasingly determines where facilities can be sited. Third, the sovereignty rationale itself remains contested among European policy researchers, who note that gigafactories depend almost entirely on non-European hardware and are yet to demonstrate a clearly justified enterprise demand case.

For BLACK& and its institutional audience, the practical implication is that "sovereign AI" in Europe is best understood not as a single infrastructure layer but as a stack — compute, energy, orchestration, and regulation — in which national projects such as the Black Sea AI Gigafactory compete for a defined pool of EU co-financing (capped at 35% of CAPEX) while carrying full exposure to grid, permitting, and offtake risk on the remainder.

1. Methodology

This paper synthesises primary and secondary sources published between January 2024 and July 2026, drawing on: (i) European Commission and EuroHPC Joint Undertaking policy documents and progress dashboards; (ii) Romanian government communications from the Ministry of Economy, Digitalisation, Entrepreneurship and Tourism and the Ministry of Energy; (iii) industry and infrastructure-market analysis from STL Partners, the Centre for European Policy Studies (CEPS), and sector publications including Datacenter Forum Romania; and (iv) contemporaneous news reporting from Romania

Insider, HotNews, and Balkan Green Energy News for verification of announcement dates and figures.

Figures reported by primary sources (EU institutions, Romanian ministries) are treated as authoritative for policy commitments; figures from secondary industry analysis are cross-checked against at least one additional source before inclusion. Where sources disagree — for example, on the total value of InvestAI's private co-investment ambitions — the range is disclosed rather than collapsed into a single figure. This is a desk-research synthesis, not a primary data-collection or econometric study; see Limitations.

Scope

The paper covers EU-level compute infrastructure policy (AI Factories, AI Gigafactories, InvestAI, the Cloud and AI Development Act) and one national case study (Romania). It does not cover foundation-model development, AI Act substantive compliance obligations, or non-EU sovereign AI programmes (e.g., UK, Gulf states, or the US CHIPS Act), which fall outside BLACK&'s stated EU/international market focus.

2. Definitions

- AI Factory — A EuroHPC-connected compute cluster built around an existing supercomputer, optimised for AI model training, fine-tuning, and startup/SME access. As of mid-2026, 19 are operational or selected across the EU.
- AI Gigafactory (AIGF) — A large-scale facility exceeding 100,000 advanced AI processors, designed for frontier-model training at a scale roughly four times that of a standard AI Factory. Up to five are planned EU-wide under InvestAI.
- InvestAI Facility — An EU financing mechanism announced in February 2025, targeting up to €200 billion in mobilised public and private capital, with €20 billion in direct public co-investment for Gigafactories.
- EuroHPC Joint Undertaking (JU) — The EU body governing high-performance computing infrastructure; its regulation was amended in 2024 (to add AI Factories) and again in January 2026 (to add Gigafactories).
- Sovereign AI — In this paper, AI infrastructure and services operated under EU or national jurisdiction, subject to EU data-protection and AI Act obligations, as distinct from AI infrastructure operated by non-EU hyperscalers.
- SPV (Special Purpose Vehicle) — The dedicated legal and governance structure proposed by Romania for the Black Sea AI Gigafactory, intended to provide neutral governance and open consortium access.
- SMR (Small Modular Reactor) — A compact nuclear reactor design; Romania's Doicești site is developed around a NuScale SMR project under a US-Romania intergovernmental cooperation agreement.

3. Analysis

3.1 The Compute Deficit and the AI Continent Action Plan

The European Commission's diagnosis, formalised in the April 2025 AI Continent Action Plan, is that Europe holds world-class research talent and a distinctive regulatory framework in the AI Act, but lacks the scale of compute, data access, and deployment that AI leadership requires. The plan is organised around five pillars: scaling compute through AI Factories and Gigafactories; building a genuine single market for data; accelerating adoption in strategic sectors; expanding the AI talent base; and supporting trustworthy AI through practical implementation of the AI Act.

By April 2026, the Commission's own one-year progress review described the "founding phase" — commissioning AI Factories, launching InvestAI, and publishing the Apply AI and Data Union strategies — as largely complete. The next phase is explicitly framed as an execution test: whether infrastructure translates into adoption, and adoption into productivity growth that justifies the investment.

3.2 From AI Factories to AI Gigafactories: Architecture and Funding

The EU's compute build-out operates in two tiers. AI Factories, funded through roughly €10 billion in combined EU and Member State investment over 2021-2027, upgrade existing EuroHPC supercomputers into AI-optimised clusters, prioritising startup and SME access. Nineteen are now operational or selected, spanning at least Finland, Germany, Greece, Italy, Luxembourg, Spain, Sweden and Romania (as a coalition partner in the Barcelona Supercomputing Centre AI Factory).

AI Gigafactories sit above this tier: each is designed to house more than 100,000 advanced processors — roughly four times the capacity of a standard AI Factory — specifically to train frontier, trillion-parameter-class models. The programme is financed through InvestAI, which the Commission aims to grow to €200 billion in combined public and private capital, of which €20 billion is direct public co-investment. An informal expression-of-interest round drew 76-77 proposals from 16 Member States, and the formal EuroHPC call for proposals — originally expected in 2025 — has been delayed twice and was, as of mid-2026, expected in Q2 2026.

3.3 The Sovereignty Debate: Hardware Dependency and Contested Value

Independent policy analysis complicates the EU's own framing. An expert roundtable convened in Amsterdam in March 2026 concluded that, absent a domestic chip and systems industry, Gigafactories risk near-total dependence on non-European hardware suppliers — a dependency the sovereignty programme is nominally designed to reduce. Participants also questioned whether European industrial demand for foundation-model training capacity is mature enough to justify a "build it and they will come" approach,

noting that most later-stage funding for European AI startups already originates from US investors.

The EU is doing Gigafactories so that politicians can say they are doing something on AI. It is a spectacle.

— Policy researcher, Amsterdam expert roundtable, March 2026, as reported by Policy Review (Verfassungsblog)

A separate, commercially oriented reading of the same facts reaches a more constructive conclusion: that Europe's earliest gigafactory-scale facilities — Microsoft/Nscale's Start Campus in Sines, Portugal (12,600 Blackwell Ultra GPUs from early 2026, scaling toward 66,000 Rubin GPUs by late 2027 on a 1.2GW-permitted site) and Deutsche Telekom/NVIDIA's Industrial AI Cloud in Munich (opened February 2026, targeting full feature parity with US hyperscaler clouds by end of 2026) — are being built by commercial actors ahead of the EU's own formal programme. On this reading, the EU's practical challenge has shifted from building sovereign compute in isolation to shaping the terms on which large-scale AI infrastructure already being built in Europe is financed, accessed, and governed.

3.4 Case Study: Romania's Black Sea AI Gigafactory

Romania submitted a formal letter of intent to EuroHPC JU in June 2025 to host a national AI Gigafactory under InvestAI, branded the Black Sea AI Gigafactory. The project is structured in two phases: Phase I at Cernavodă, drawing on the site's existing nuclear capacity (Units 1-2, approximately 1,400 MW combined) and planned Units 3-4, alongside direct fibre and submarine-cable connectivity to major European nodes; and Phase II at Doicești, an industrial site earmarked for a NuScale small modular reactor (approximately 462 MW), developed under a US-Romania intergovernmental cooperation agreement, with potential for hybrid cooling and co-location.

The combined project targets more than 100,000 AI accelerators powered by up to 1,500 MW of predominantly low-carbon nuclear and renewable capacity, at an estimated investment of €4-5 billion, with a target operational date at the end of 2028. Romania's government has stated that EU and national public financing can cover up to 35% of total CAPEX, with the remainder to be mobilised through private partners, Romania's national development bank, and institutional investors — the World Bank has reportedly expressed interest following meetings with Romanian officials in the United States.

The Romanian bid is explicitly framed in regional as well as national terms, positioning the facility as infrastructure for the Republic of Moldova and other Black Sea states, including Ukraine and Turkey, alongside a competing Dutch proposal that emphasises direct access to North Sea wind generation. By June 2026, the government had moved from letter-of-intent to launching an Expression of Interest to select a consortium leader, while reporting that prospective investors were seeking clearer state guarantees, co-financing terms, and risk-mitigation mechanisms before committing capital — a

sequencing point of direct relevance to any BLACK& engagement in the consortium-formation process.

3.5 Energy as the Binding Constraint

Across every facility examined in this paper, siting decisions are driven first by power availability, not chip allocation. European industrial electricity prices run roughly double US levels, and while electricity is a single-digit share of total model-training cost, that differential compounds across training runs costing hundreds of millions of euros. Goldman Sachs Research forecasts global data-centre electricity demand rising 50% by 2027 and 165% by the end of the decade relative to 2023, a trajectory that makes grid capacity, permitting speed, and firm power-purchase agreements the effective gating factor for new AI infrastructure — a dynamic visible in Nebius's parallel commitments to 310 MW (Finland) and 240 MW (Lille, France) facilities, each at a similar order of magnitude to the EU's own planned Gigafactories.

3.6 Regulatory Layer: The AI Act and the Cloud and AI Development Act

Two regulatory instruments intersect with the infrastructure build-out. The EU AI Act's high-risk obligations begin entering into force from 2 August 2026, with some obligations reportedly at risk of slipping into 2027 pending a political simplification agreement reached in May 2026. Separately, the proposed Cloud and AI Development Act (CADA) aims to reduce strategic dependency on non-EU cloud providers and is intended to triple EU data-centre capacity over five to seven years. For enterprises and investors, GAIA-X certification is increasingly cited as the practical procurement benchmark for verifying EU-compliant, sovereignty-conscious AI services, particularly in regulated-industry RFPs.

4. Data and Charts

The following exhibits summarise the quantitative core of this analysis: the scale of Europe's compute layer, the capital committed across programmes, the specific power foundation of the Romanian case study, and the sequencing of policy milestones referenced throughout Section 3.

Exhibit 1. Europe's sovereign compute layer by facility count and typical scale, 2026.

Exhibit 2. Capital mobilised across EU and national AI infrastructure programmes (EUR, billions). The InvestAI figure reflects the Commission's combined public-and-private mobilisation target, not a confirmed disbursement.

Exhibit 3. Installed and planned power capacity underpinning the Black Sea AI Gigafactory's two phases.

Exhibit 4. Sequencing of EU and Romanian sovereign AI infrastructure milestones, 2024-2026.

5. Strategic Implications

For institutional and government audiences assessing exposure to this build-out, three implications follow directly from the evidence above.

- Sovereign AI is a stack, not a facility. Compute (Factories/Gigafactories), energy (nuclear, SMR, renewables, grid), orchestration (federated cloud initiatives such as EURO-3C), and regulation (AI Act, CADA, GAIA-X) move on different timelines and different risk profiles; treating a single Gigafactory announcement as the investable unit understates both the opportunity and the execution risk.
- Public co-financing is a minority stake in the capital structure. With EU and national public financing capped at roughly 35% of CAPEX for facilities such as the Black Sea AI Gigafactory, the majority of capital risk sits with private partners, development banks, and institutional investors — precisely the audience already seeking clearer state guarantees before committing.
- Early-mover commercial facilities are setting de facto standards ahead of the EU's own formal programme. Enterprises and investors evaluating sovereign AI exposure in 2026 are, in practice, choosing between operational commercial sovereign clouds (Germany, Portugal, France) and pre-operational public-programme bids (including Romania's) that carry longer timelines but potentially deeper regional and regulatory alignment.

6. Limitations

- This is a desk-research synthesis of public policy documents, industry analysis, and news reporting current to July 2026; it does not incorporate primary interviews, procurement data, or non-public consortium financials.
- Key figures — notably InvestAI's €200 billion mobilisation target and the €176 billion cumulative EU data-centre investment estimate for 2026-2031 — are forward-looking projections from EU and industry sources, not confirmed disbursements, and should be treated as directional rather than exact.
- The EuroHPC formal call for Gigafactory proposals had not been published at the time of writing; the competitive outcome for Romania's bid, and the final capital structure of the Black Sea AI Gigafactory, remain undetermined.
- The AI Act enforcement timeline referenced in Section 3.6 was subject to an active political simplification process as of May-June 2026; specific obligation dates may shift after publication.
- This paper does not independently verify electricity-price differentials, energy-demand forecasts, or vendor GPU-count disclosures beyond the cited secondary sources; figures from industry analysts (e.g., Goldman Sachs Research, STL Partners) are reported as attributed estimates, not independently audited data.

- Geographic scope is limited to the EU and Romania; comparative sovereign-AI programmes in the UK, Gulf states, and the United States are outside the scope of this paper.

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8. Revision History

Version	Date	Notes
0.1	01 Jul 2026	Internal draft — outline and source collection
1.0	15 Jul 2026	Initial publication — pilot paper for the BLACK& Research series

Next scheduled review: upon publication of the EuroHPC formal Gigafactory call for proposals, or Q4 2026, whichever is earlier.

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